DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL
FINAL EXAMINATION
Course Title: Data Structures
Course Code: CSE-1201
2nd Semester
Session: 2017-18

5 set

Time: 3 Hours **Marks: 60**

1. a) Write down the definition of algorithms as technology. For input of size m, insertion sort runs in $8n^2$ steps, while merge sort runs in $64n \log n$ steps. For which values of n does insertion sort beat merge sort? [4]

   b) Insertion sort can be expressed as a recursive procedure as follows. In order to sort A [1 . . n] , we recursively sort A [1 . . n − 1] and then insert A[n] into the sorted array A [1 . . n − 1] . Write a recurrence for the running time of this recursive version of insertion sort. [4]

   c) Analyze the algorithm and find out the best case and worst case running time. [4]

```
INSERTION-SORT(A)
for j ← 2 to n
    do key ← A[j]
       ▷ Insert A[j] into the sorted sequence A[1 .. j − 1].
       i ← j − 1
       while i > 0 and A[i] > key
           do A[i + 1] ← A[i]
              i ← i − 1
       A[i + 1] ← key
```

2. a) Prove that the running time of an algorithm is $\Theta(g(n))$ if and only if its worst-case running time is $O(g(n))$ and its best-case running time is $\Omega(g(n))$. [2]

   b) Define divide and conquer. Solve the problem, 5^{31} % 9 using divide and conquer technique. [3]

   c) Write down the operation of HEAPSORT on the array A = < 5, 13, 2, 25, 7, 17, 20, 8, 4> [4]

   d) What is the running time of QUICKSORT when all elements of array A have the same value? [3]

3. a) Write down the recursive formula of the "*Tower of Hanoi*" problem. [2]

   b) Write down the O(n+k) complexity algorithm for sorting an array. [3]

   c) Define Stack and Queue. Given the following numbers 10, 4, 6, 8, 9 and 3, write [3]
      i. What would be the value to the stack, if three times push, one time pop, one time push and one time pop. Where push and pop operation refers to insert and pull an element to the stack respectively.
      ii. What would be the value to the queue, if two times enqueue, one time dequeue, three times enqueue and four times dequeue.
      iii. Write down the operations for making empty stack to the numbers.

   d) Define Linked list. Implement a stack using a singly linked list L . The operations PUSH and POP should still take O(1) time. [4]

4. a) Demonstrate what happens when we insert the keys 5, 28, 19, 15, 20, 33, 12, 17, 10 into a hash table with collisions resolved by chaining. Let the table have 9 slots, and let the hash function be h(k) = k mod 9. [3]

   b) What do you mean by dynamic programming? Write down the program to find out the Longest Common Subsequence on the sequences X= <A, B, C, B, D, A, B> and Y= <B, D, C, A, B, A>. [3]

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c) Is disjoint set a data structure? Justify your answer. Suppose that CONNECTED-COMPONENTS is run on the undirected graph $G = (V, E)$, where $V = \{a, b, c, d, e, f, g, h, i, j, k\}$ and the edges of E are processed in the following order: (d, i), (f, k), (g, i), (b, g), (a, h), (i, j), (d, k), (b, j), (d, f), (g, j), (a, e), (i, d). Now, [3]

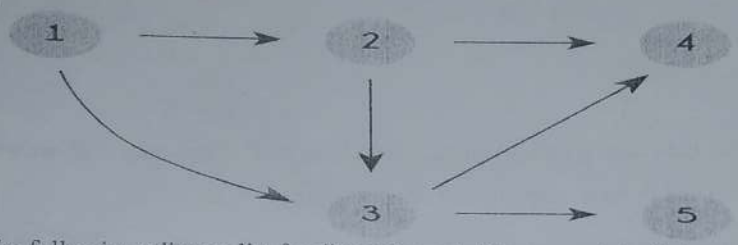
i. What do you mean by disjoint set data structure?

ii. Find out the connected components from the following graph.

iii. Also prove that after all edges are processed by CONNECTED-COMPONENTS, two vertices are in the same connected component if and only if they are in the same set.

d) What is the running time of BFS if we represent its input graph by an adjacency matrix and modify the algorithm to handle this form of input? [3]

5. a) A **topological sort** or **topological** ordering of a directed graph is a linear ordering of its vertices such that for every directed edge **u** and **v** from vertex **u** to vertex **v**, **u** comes before **v** in the ordering. Now using the graph, Find out the topological ordering of the graph using DFS algorithm. [4]



b) Consider the following adjacent list for directed graph. Now your tasks is to, [4]

adj(y) = [x],
adj(x) = [z],
adj(z) = [y,w],
adj(w) = [x],
adj(s) = [z,w],
adj(v) = {s,w],
adj(t) = [u,v],
adj(u) = [v].

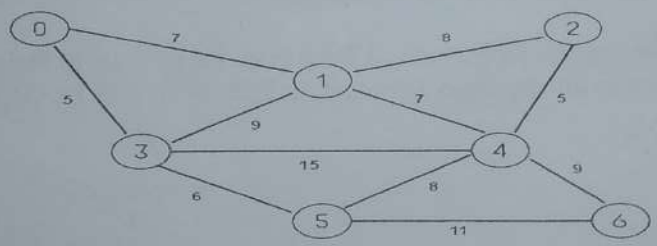
i. Construct a graph from the above information.

ii. Identify the tree edge, back edge, cross edge and forward edge for the graph.

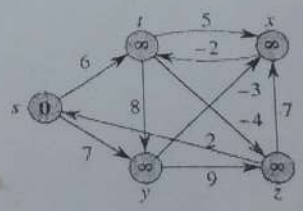
c) Professor Bacon claims that the algorithm for strongly connected components would be simpler if it used the original (instead of the transpose) graph in the second depth-first search and scanned the vertices in order of increasing finishing times. Does the simpler algorithm always produce correct result? [4]

6. a) Write down the kruskal's algorithm for minimum spanning tree. [3]

b) Define safe edge. Write down the execution of Prim's algorithm on the following graph. [4]



c) Run the Bellman-Ford algorithm on the directed graph of following figure (Where initialization single source is done by source 's'), using vertex z as the source. In each pass, relax edges in the same order as in the figure, and show the d and π values after each pass. Now, change the weight of edge (z, x) to 4 and run the algorithm again, using s as the source. [5]



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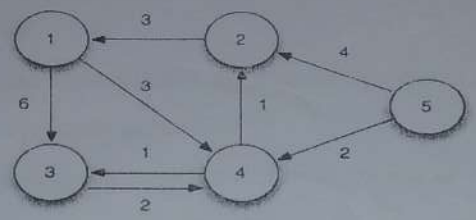
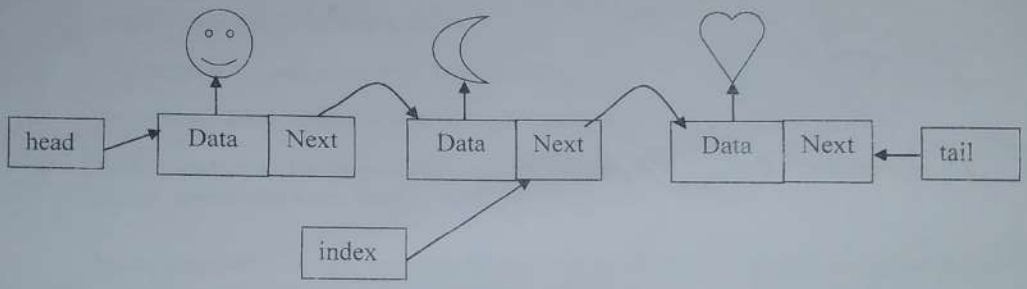
7. a) Suppose that x, y, and z are strings such that $x \sqsupset z$ *and* $y \sqsupset z$. Prove that if $|x| \leq |y|$ *then* $x \sqsupset y$, *if* $|x| \geq |y|$ *then* $y \sqsupset x$ *and if* $|x| = |y|$ *then* $x = y$. [2]

b) For the following text **T=abababababa** and pattern **P=ababa,** Construct the string matching automaton and find out the number of hits does the following string matching algorithms encounter to the text. [3]

c) Run the Dijkstra's algorithm on the following graph using node '1' as the source in order to find single source shortest path. [4]



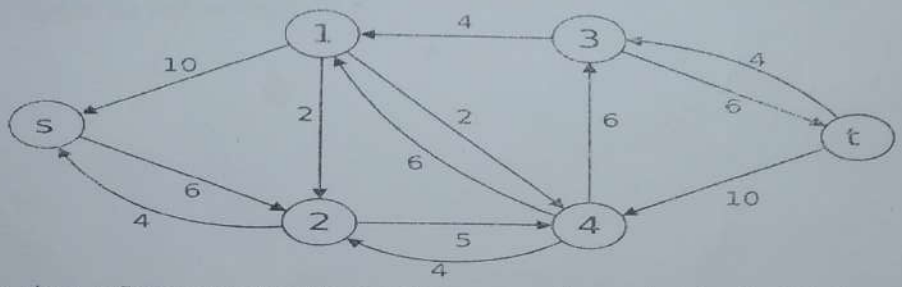
d) Consider the Linked List shown below. Please write a sequence of lines of code that will remove the node named by "tail". **You do not need to write a generic method – just use any programming language to describe the deletion step-by-step.** [3]



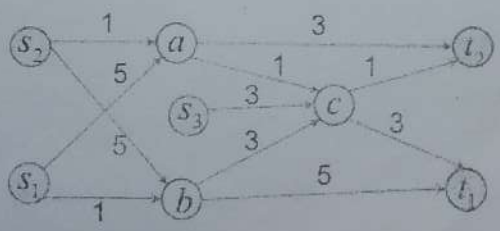
8. a) Define flow network and anti-parallel edge with example. Also write down the properties of the flow network. [3]

b) Write down the algorithm of Edmond-Karp and when we use the algorithm with appropiate example. [3]

c) Implement the ford-fulkerson algorithm for the graph below from source node 's' to sink node 't'. [4]



d) We are given a flow network with several sources and several sinks. Explain how to use an algorithm for finding a maximum flow (in a standard flow network) for this case. [2]



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